

Hands-On 10 Project: HO10_Unit_Testing

Commands:

```
ng new HO10_Unit_Testing --routing --style=css
cd HO10_Unit_Testing
ng g c components/course-card
ng g s services/course
ng test
```

What we did:

Created unit tests for component and service. All tests passed.

course-card.ts

```
import { Component, Input, Output, EventEmitter, OnChanges, SimpleChanges } from
 '@angular/core';
```

```
@Component({
  selector: 'app-course-card',
  standalone: true,
  imports: [],
  templateUrl: './course-card.html',
  styleUrls: ['./course-card.css']
})
export class CourseCard implements OnChanges {
  @Input() course: any;
  @Output() enrollRequested = new EventEmitter<number>();

  ngOnChanges(changes: SimpleChanges) {
    console.log('Course Changed:', changes);
  }

  enroll() {
    this.enrollRequested.emit(this.course.id);
  }
}
```

course-card.html

```
<div>
  <h3>{{ course.name }}</h3>
  <p>Course Code: {{ course.code }}</p>
  <p>Credits: {{ course.credits }}</p>
  <button (click)="enroll()">Enroll</button>
</div>
```

course-card.spec.ts

```
import { ComponentFixture, TestBed } from '@angular/core/testing';
import { By } from '@angular/platform-browser';
```

```

import { vi } from 'vitest';
import { CourseCard } from './course-card';

describe('CourseCard', () => {
  let component: CourseCard;
  let fixture: ComponentFixture<CourseCard>;

  beforeEach(async () => {
    await TestBed.configureTestingModule({
      imports: [CourseCard]
    }).compileComponents();

    fixture = TestBed.createComponent(CourseCard);
    component = fixture.componentInstance;
  });

  it('should create', () => {
    component.course = { id: 1, name: 'Data Structures', code: 'CS101', credits: 4 };
    fixture.detectChanges();
    expect(component).toBeTruthy();
  });

  it('should render course name', () => {
    component.course = { id: 1, name: 'Data Structures', code: 'CS101', credits: 4 };
    fixture.detectChanges();

    const h3 = fixture.debugElement.query(By.css('h3')).nativeElement;
    expect(h3.textContent).toContain('Data Structures');
  });

  it('should emit enroll event', () => {
    component.course = { id: 1, name: 'Data Structures', code: 'CS101', credits: 4 };
    fixture.detectChanges();

    const emitSpy = vi.spyOn(component.enrollRequested, 'emit');

    fixture.debugElement.query(By.css('button')).nativeElement.click();

    expect(emitSpy).toHaveBeenCalledTimes(1);
  });
});

```

course.service.ts

```

import { Injectable } from '@angular/core';

@Injectable({
  providedIn: 'root'
})

```

```
export class CourseService {  
  getCourses() {  
    return [  
      { id: 1, name: 'Angular' },  
      { id: 2, name: 'React' }  
    ];  
  }  
}
```

course.spec.ts

```
import { TestBed } from '@angular/core/testing';  
import { CourseService } from './course.service';
```

```
describe('CourseService', () => {  
  let service: CourseService;
```

```
  beforeEach(() => {  
    TestBed.configureTestingModule({});  
    service = TestBed.inject(CourseService);  
  });
```

```
  it('should create service', () => {  
    expect(service).toBeTruthy();  
  });
```

```
  it('should return courses', () => {  
    expect(service.getCourses().length).toBeGreaterThan(0);  
  });  
});
```

Output screenshot:

Test Files 13 passed
Tests 16 passed
PASS

